KAHBOD AEINI

Electrical and Computer Engineering Department, University of Waterloo, Waterloo, Ontario, Canada

kaeini@uwaterloo.ca — LinkedIn — GitHub — Website — Google Scholar

EDUCATION

University of Waterloo Jan 2026 - Ongoing
PhD in Electrical and Computer Engineering
Supervisor: Prof. Wojciech Golab

University of Calgary 2023 - 2025
Master of Science in Computer Science
Supervisor: Prof. Philipp Woelfel
Thesis: *A Space-Optimal Randomized Wait-Free Lock*

Sharif University of Technology 2019 - 2023
Bachelor of Computer Engineering
Thesis: *Fraud Detection from Sequences of Bank Transactions Using Various Machine Learning Algorithms*

National Organization for Development of Exceptional Talents 2013 - 2019
High School Diploma in Mathematics and Physics

RESEARCH INTERESTS

Distributed Systems, Concurrent & Randomized Algorithms, Persistent Memory & Recoverable Objects, Shared-Memory Systems, Multicore Programming, Advanced Data Structures, Database Systems

PUBLICATIONS

Kahbod Aeini, Dante Bencivenga, George Giakkoupis, Philipp Woelfel
Simple and Efficient Randomized Wait-Free Locks.
45th ACM Symposium on Principles of Distributed Computing (PODC), 2026. **Best Paper Award.**

Kahbod Aeini, Wojciech Golab
Analyzing Linearizability in Relativistic Distributed Systems.
8th ApPLIED Workshop, co-located with PODC 2026, 2026.

Kahbod Aeini.
A Space-Optimal Randomized Wait-Free Lock.
Master's Thesis, University of Calgary, 2025.

WORK EXPERIENCE

Software Engineer, ArcTrade (part-time) Dec 2025 - Ongoing
- Developing backend services and APIs for ArcTrade's retail energy technology platform.
- Implementing secure, automated electronic data interchange (EDI) with external trading partners, enabling encrypted and verifiable exchange of business transactions.
- Contributing to components supporting energy trading, risk management, pricing, billing and settlement workflows in North American electricity markets.
- Collaborating with cross-functional teams to design scalable and efficient software solutions.

Data Engineer, AI Lida Feb 2022 - Sep 2022
- Developed, maintained and optimized data pipelines for ETL processes using Python and PostgreSQL.
- Collaborated with the security department to gather requirements and deliver data solutions that supported business objectives.
- Interviewed, hired, trained and supervised a team of 5 data engineer interns, providing mentorship and performance feedback.

RESEARCH EXPERIENCE

PhD Student, Research Assistant, Distributed Algorithms Jan 2026 - Ongoing
Advisor: Prof. Wojciech Golab, University of Waterloo
Conducting research on persistent memory, shared-memory distributed systems, and recoverable objects, focusing on the design and analysis of efficient algorithms for synchronization, consensus, and fault tolerance in distributed systems.

Master's Student, Research Assistant, Distributed Algorithms Jun 2024 - Jun 2025
Supervisor: Prof. Philipp Woelfel, University of Calgary
Designed and analyzed randomized wait-free lock algorithms achieving optimal trade-offs between space and time complexity in shared memory systems, including the first space-optimal algorithm and complementary time-optimal variants.

Bioinformatics, Research Assistant (Volunteer, Remote) Feb 2023 - Aug 2023
Supervisor: Prof. Mehdi Pirooznia, Johns Hopkins University
Conducted feature selection and developed machine learning models to predict immune cell composition in lung cancer. Performed identification of immune cell types, analysis of differentially expressed genes, functional enrichment analysis, and classification of cancer subtypes using machine learning techniques.

Deep Learning, Research Assistant, Machine Learning Oct 2021 - Feb 2022
Supervisor: Dr. Fatemeh Baharifar & Dr. Vahid Motaghed, *IPM*
Developed and evaluated face clustering algorithms using graph convolutional networks (GCN) for linkage prediction and pairwise classification approaches.

AWARDS AND SCHOLARSHIPS

Best Paper Award, 45th ACM Symposium on Principles of Distributed Computing (PODC) 2026
Awarded for *Simple and Efficient Randomized Wait-Free Locks*

Departmental Research Award, Department of Computer Science, University of Calgary 2023
Won the RA award, which is granted to the best students in each department to step away from TA responsibilities and focus on their research

Top 100 Students in Country 2019
Ranked 94 (31st in the Region) in National Universities Entrance Exam among around 160,000 participants

National Elites Fellowship 2019
Fellowship in Iran's National Elites Foundation

TEACHING EXPERIENCE

University of Calgary 2023 - 2025
Led and managed TA teams across undergraduate and graduate courses (up to 300 students total), including Theoretical Foundations of Computer Science (7 TAs, 150+ students), Randomized Algorithms (graduate), Programming Paradigms (Haskell & Prolog), and Python Programming.

- **Theoretical Foundations of Computer Science II (Winter 2024 & 2025 & 2026)** — Coordinated 7 TAs, supported 150+ students in automata theory and probability.
- **Programming Paradigms (Fall 2025)** — Managed 6 TAs, organized and held tutorials in Haskell & Prolog for 75+ students.
- **Randomized Algorithms (Graduate, Fall 2024)** — Delivered tutorials on randomized data structures, lower bound techniques, and complexity classes.
- **Python Programming (Fall 2023)** — Taught fundamentals of Python, data structures, and algorithms for 50+ students.

Sharif University of Technology 2020 - 2023
Served as Head TA across multiple undergraduate courses (TA teams of 10-25, 200+ students), including Data Structures & Algorithms, Numerical Calculations, and Mobile Programming.

- **Data Structures and Algorithms** — Designed assignments & projects, led TA group of 25+.
- **Numerical Calculations** — Developed final project curriculum, led TA group of 10+.
- **Mobile Programming** — Designed and graded Android development projects.

PROFESSIONAL SERVICE

External Reviewer, International Symposium on Distributed Computing (DISC) 2026

NOTABLE PROJECTS

Concurrent Data Structures on Multicore Hardware. Designed, implemented, and benchmarked TLE-based concurrent binary search trees and expandable probing hash tables using Intel Restricted Transactional Memory (RTM) on a 192-thread NUMA server. Analyzed transactional commit/abort behavior, false sharing, NUMA effects, and memory reclamation strategies (EBR) using `perf`; explored fallback-path parallel expansion with OpenMP and CAS. (CS 798: Advanced Research Topics in Multicore Programming, Winter 2026)

Fraud Detection Model. As the Bachelor thesis, titled "Fraud Detection from Sequences of Bank Transactions Using Various Machine Learning Algorithms", I developed, trained and tested three different Machine Learning models for Fraud Detection on a dataset of banking transactions. The methods then were analysed and compared to each other. (Git - Jun 2023)

Smart Water Supplier. Contributing to a team of four in designing and implementing a smart water container system, integrating hardware components like Raspberry Pi, proximity sensor, and water level sensors, coupled with a Django-based web app. (Git - Jun 2023)

Information Retrieval System. Designed and implemented a retrieval system for twitter interaction and health articles with GUI, Classification and Clustering tweets and articles. (Git - Jun 2022)

RELEVANT COURSES

Multicore Programming (Advanced Research Topics): A+ - **Randomized Algorithm**: A+ - **Algorithms for Distributed Computing**: A+ - **Data Mining & Machine Learning**: A - **Advanced Information Retrieval**: A+ - **Database Design**: A - **Design of Algorithms**: A - **Artificial Intelligence**: A+ - **Data Structures and Algorithms**: A+ - **Linear Algebra**: A+ - **Numerical Computation**: A+ - **Real Time Systems**: A

SKILLS

Languages: C#, Python, R, Haskell, Java, C/C++, Prolog

Frameworks & Tools: .NET / ASP.NET Core, Entity Framework, Docker, Git, Django, PyTorch, TensorFlow, scikit-learn, Numpy, Pandas, OpenCV, Android, limma

Databases: Microsoft SQL Server, PostgreSQL, MongoDB, MySQL

Other: Linux, NixOS, Kali